Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	d77520d5-4b32-4fd9-a10c-33a16b60265f
Date	2026-11-01 18:08:54
Status	Reviewed by expert (expert)

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Tool Wear and Die Damage

Reason: The presence of scratches, dents, and dimensional shifts (L2 and H1 deviations) strongly correlates with progressive wear of punch and die tooling, which alters geometry and damages part surfaces [587aa665-c6f1-48df-8783-09d65ff4f82a].

Effect: Deformation, Dimensional Inaccuracy

Confidence: 76%

Supporting evidence
- **587aa665-c6f1-48df-8783-09d65ff4f82a**: Tool Wear / Die Damage Root Cause Analysis Defect Types: scratch, dent, edge burr, dimensional shift Process: bending, stamping, forming Root Cause: Progressive wear of punch/die tooling causes dimensional drift, surface scratches, edge burrs and dents on formed parts. Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces, edge burrs from worn cutting edges,

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake
2. Verify material lot

Preventive actions (stop it recurring)

1. Schedule monthly calibration
2. Incoming inspection

Precautions

- Ensure tooling alignment is checked after every regrind

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Improper Bending Force	67%	The detected major deformation and dimensional inaccuracies in the bent features are consistent with improper force settings or exceeding the material's elastic limit during the bending process [586f086d-0deb-40c2-a71d-f840aaa55a60].
3	Material Springback	38%	Dimensional inaccuracies in bent features and overall deformation are frequently caused by the elastic recovery of SPCC steel after the bending force is released, leading to parts falling out of tolerance [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.